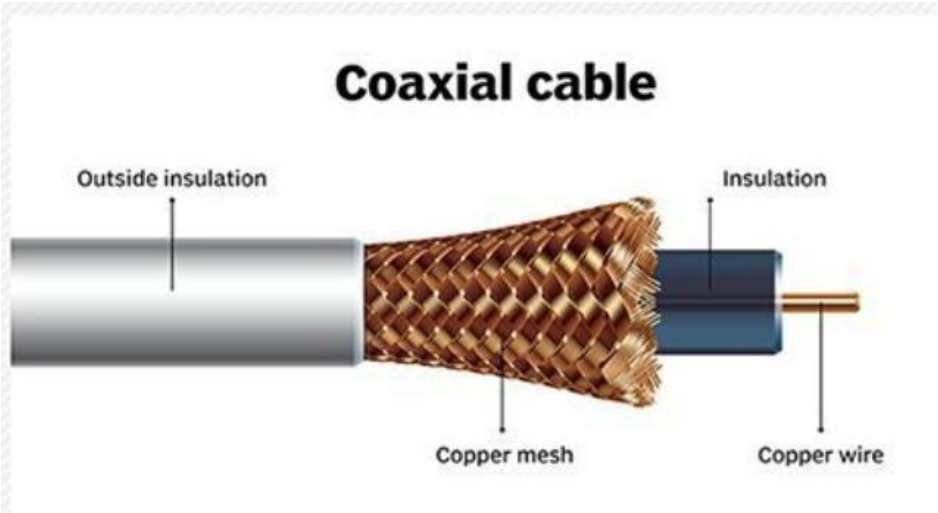
# Practical 01

## Practical 1: Identifying Network Component

a. **Aim:** Identify and document types of cables (Coaxial, Twisted Pair, Fiber Optic)

### Coaxial Cable:

Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable. Ithas a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.




### Applications:

- Coaxial cable was widely used for both analog and digital data transmissions.
- It has higher bandwidth.
- Inexpensive when compared to fiber optical cables.
- It uses for longer distances at higher data rates.
- Excellent noise immunity.
- Used in LAN and Television distribution.

### Disadvantage :

- Single cable failure can fail the entire network.
- Difficult to install and expensive when compared with twisted pairs.
- If the shield is imperfect, it can lead to grounded loop.

## Twisted pair cable:

A twisted pair cable is a type of cable made by putting two separate insulated wires together in a twisted pattern and running them parallel to each other. This type of cable is widely used in different kinds of data and voice infrastructures.

## Twisted pair is of two types:

1. Shielded Twisted Pair(STP)
2. Unshielded Twisted Pair(UTP)

## Shielded Twisted Pair:

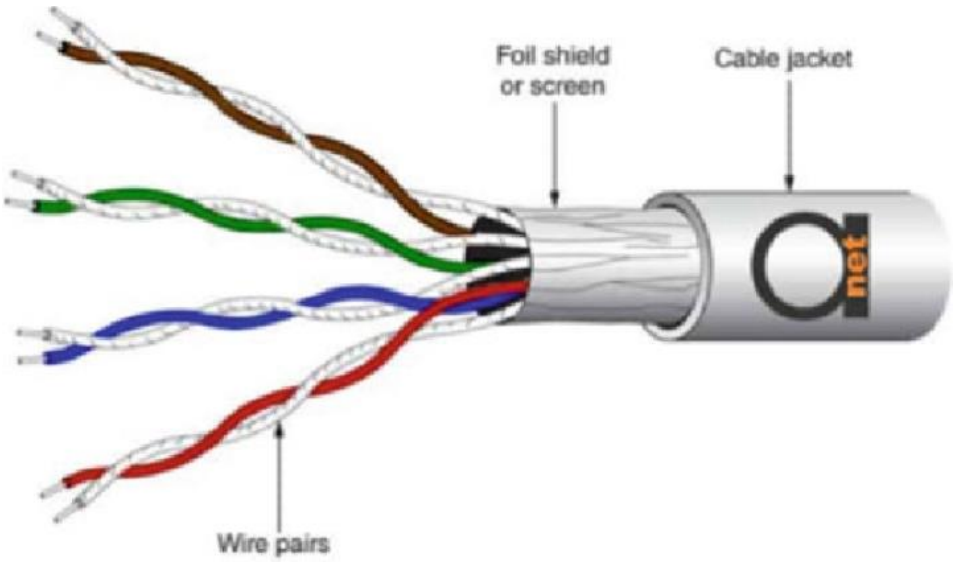
Shielded Twisted Pair (STP) cables additionally have an overall conducting metallic shields covering four twisted pair wires. There may be another conducting metallic shields covering individual twisted pairs also. These metallic shields blocks out electromagnetic interference to prevent unwanted noise from the communication circuit.

## Advantage of Shielded Twisted Pair:

1. The cost of the shielded twisted pair cable is not very high and not very low.
2. An installation of STP is easy.
3. It has higher capacity as compared to unshielded twisted pair cable.
4. It has a higher attenuation
5. It is shielded that provides the higher data transmission rate.

## Disadvantages:

1. It is more expensive as compared to UTP and coaxial cable.
2. It has a higher attenuation rate.



## Unshielded Twisted Pair(UTP)

An unshielded twisted pair is widely used in telecommunication. It is most common type when compared with shielded twisted pair cable which consists of two conductors usually copper, each with Its own colour plastic insulator

## Advantages Of Unshielded Twisted Pair:

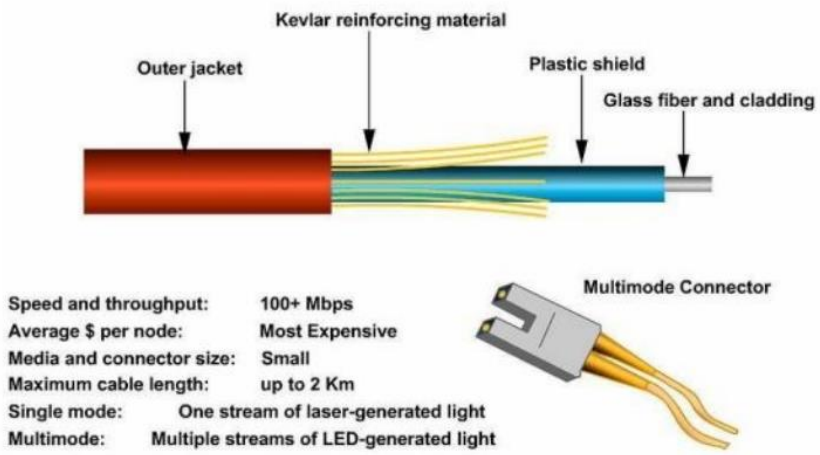
1. It is cheap.
2. Installation of the unshielded twisted pair is easy.
3. It can be used for high-speed LAN.

**Disadvantage:**

1. The cable can only be used for shorter distances

**Fiber Optic Cable:**

A fiber optic cable is a network cable that contains strands of glass fibers inside an insulated casing. They're designed for long-distance, high-performance data networking, and telecommunications. compared to wired cables, fiber optic cables provide higher bandwidth and transmit data over longer distances. Fiber optic cables support much of the world's internet, cable television, and telephone systems.



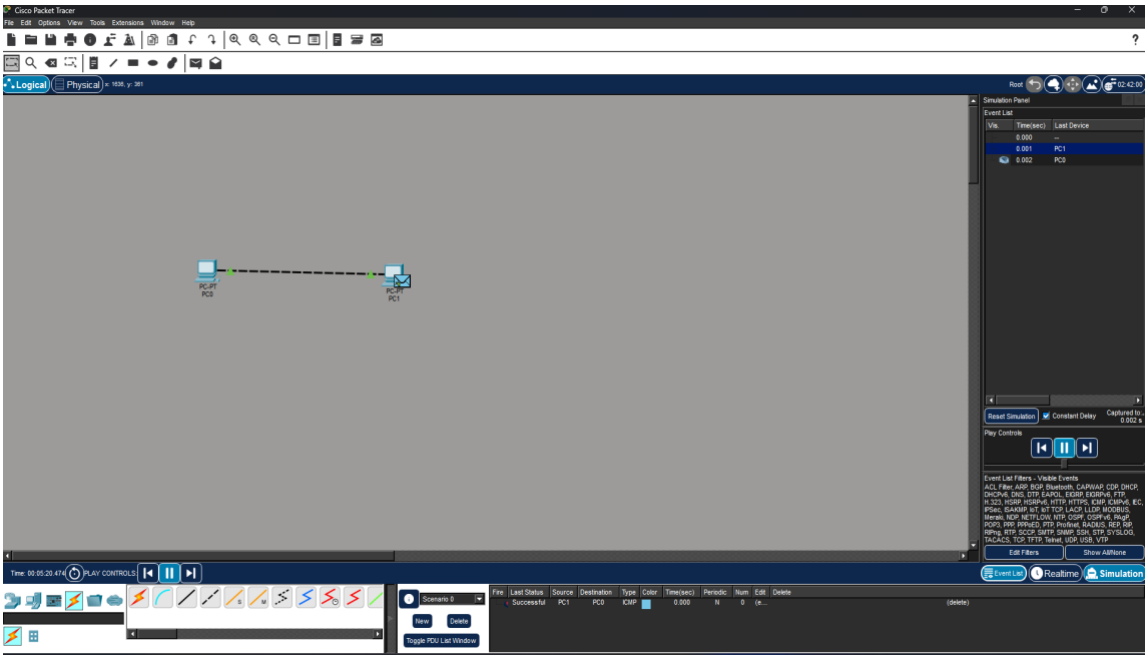

## Advantages of Fiber Optic Cables:

1. The loss of signal in optical fiber is less than that in copper wire.
2. Opticalfibers usually have a longer life cycle for over 100 years.

## Disadvantage:

1. It is expensive.
2. Difficult to install.

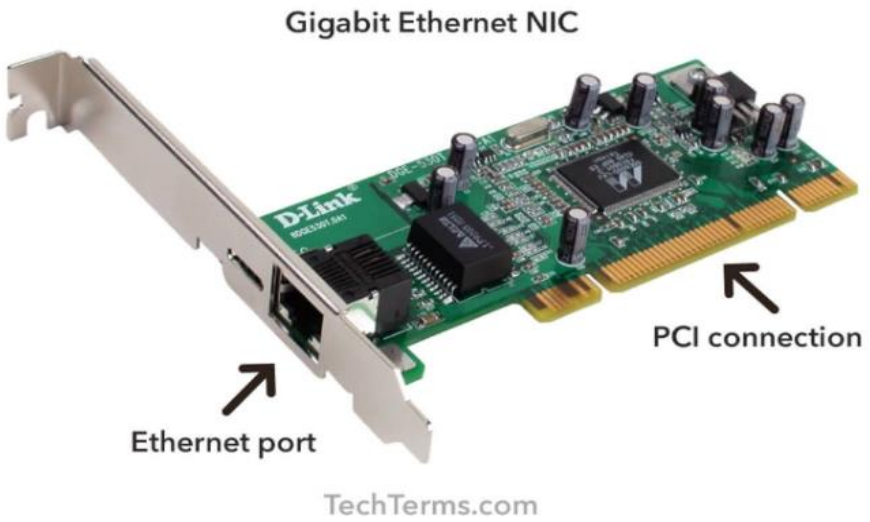
b. **AIM:** Demonstrate connecting PCs using a switch and straight-through cables.



c. **AIM:** Examine functions of NIC, hub, router, and switch physically or virtually.

## Network Interface Card (NIC)

A hardware component inside a computer that connects it to a network.




## Functions:

1. Provides a physical connection to the network (wired or wireless)
2. Assigns a unique MAC address to the device
3. Converts data into signals (electrical or wireless) for transmission
4. Receives incoming data and passes it to the system

## Hub:

A basic networking device that connects multiple computers.

**Functions**:

1. Receives data from one device
2. Broadcasts the data to all connected devices
3. Works at the Physical Layer (Layer 1)

## Limitations:

- No intelligence (cannot filter data)
- Causes network congestion

## Switch:

A switch is a Networking device in a computer network that connects other devices together. Multiple data cables are plugged into a switch to enable communication between different networked Devices. switches manage the flow of data across a network by transmitting a received network packet Only to the one or more devices for which the packet is intended

## Functions:

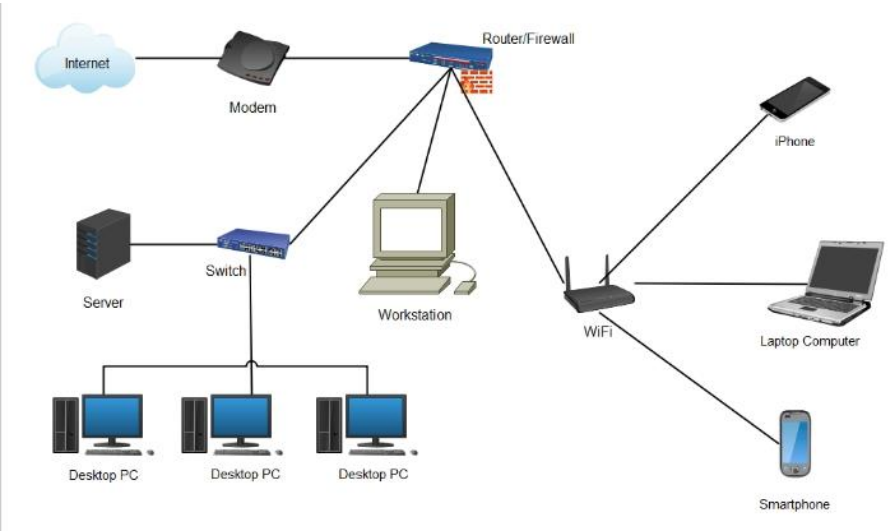
1. Sends data only to the intended device
2. Uses MAC addresses to identify devices
3. Reduces network traffic
4. Works at the Data Link Layer (Layer 2)

## Advantages over hub:

- Faster and more efficient
- Avoids unnecessary data transmission

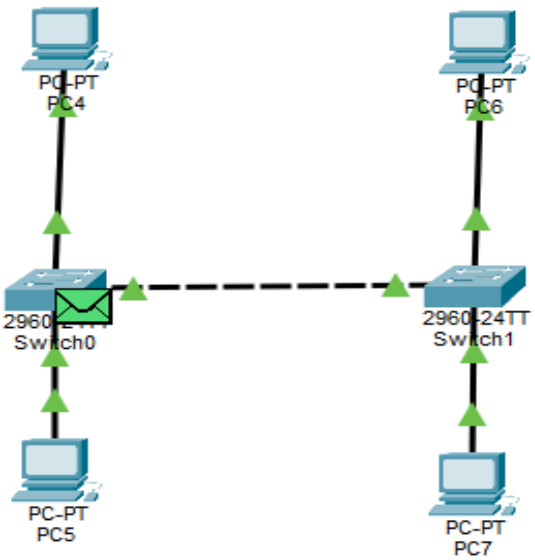
## Router:

A router is an electronic device that interconnects two or more computer networks, and Selectively interchanges packets of data between them. Each data packet contains address Information that a router can use to determine if the source and destination are on the same network, Or if the data packet must be transferred from one network to another. Where multiple routers are Used in a large collection of interconnected networks, the routers exchange information about target System addresses, so that each router can build up a table showing the preferred paths between any two systems on the interconnected networks.

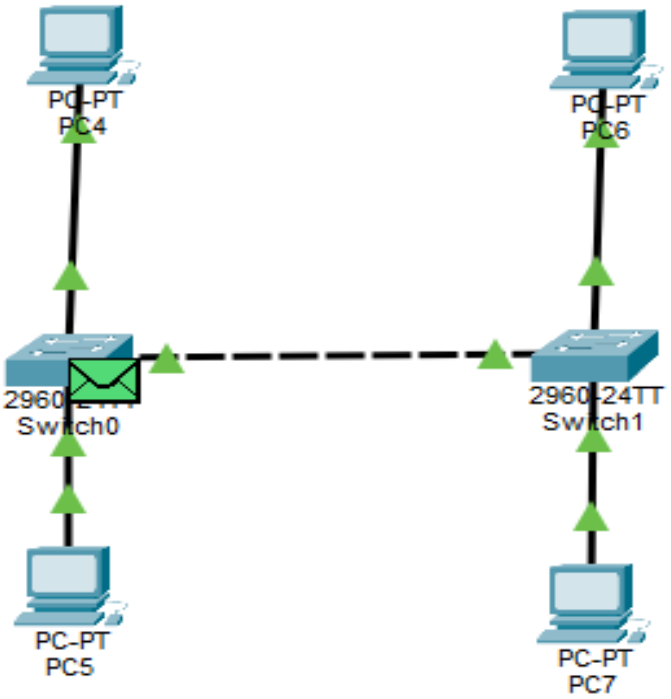


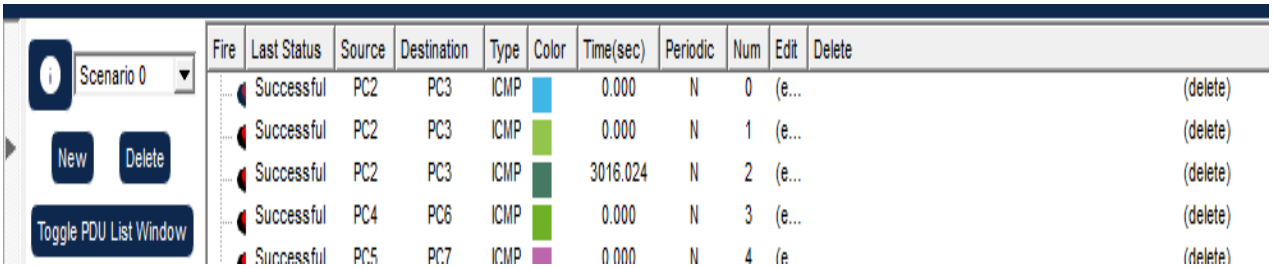

## Functions:

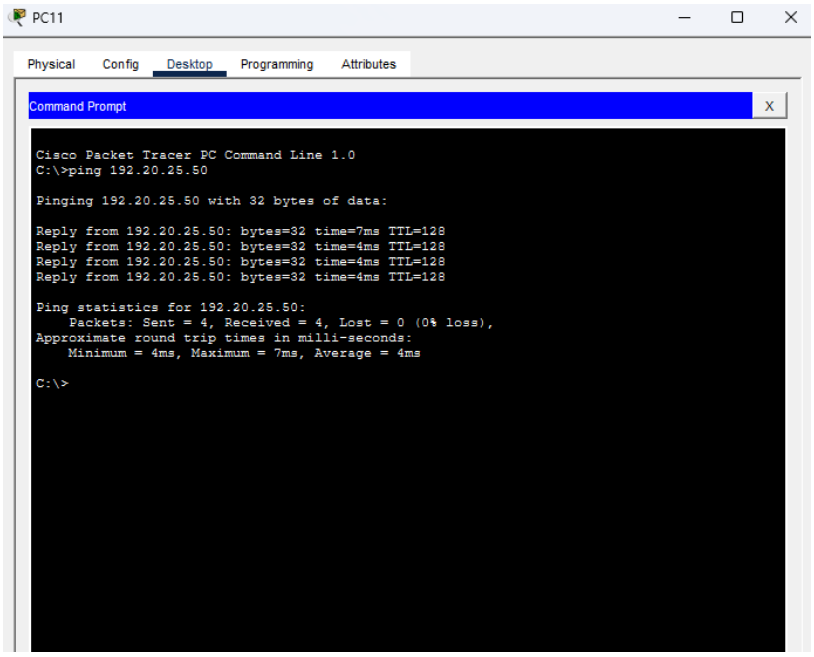
1. Routes data between networks using IP addresses
2. Connects LAN to WAN (Internet)
3. Provides Wi-Fi connectivity
4. Performs NAT (Network Address Translation)



**d. AIM:** Colour code for crimping LAN (Cat 5/6/7) cable



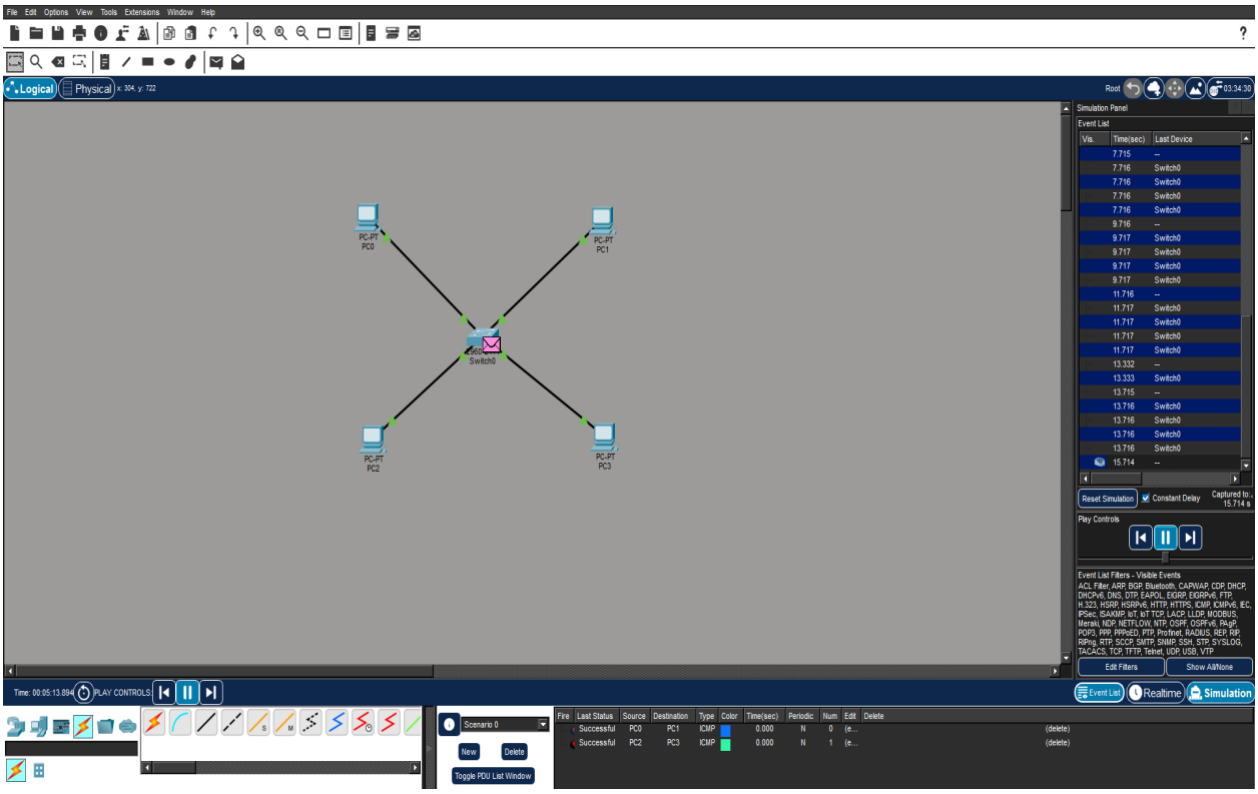



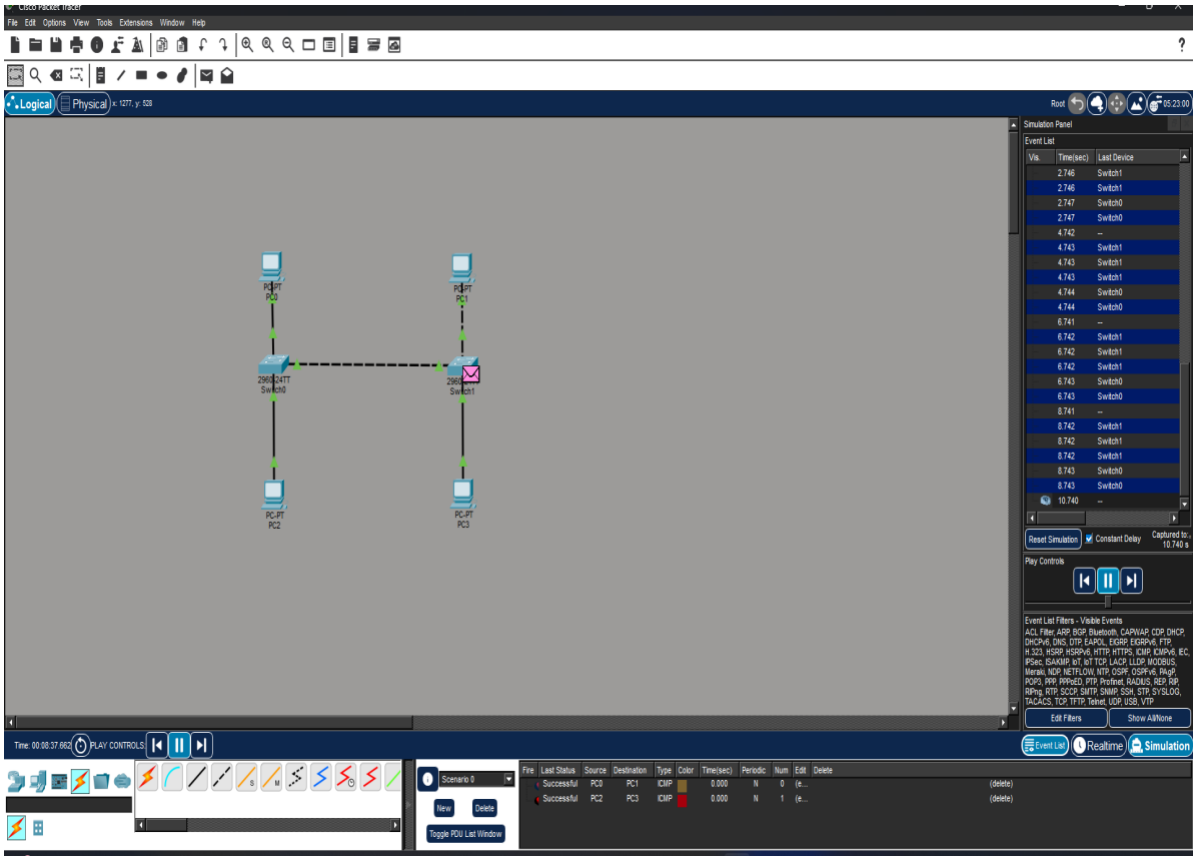
# Practical 02

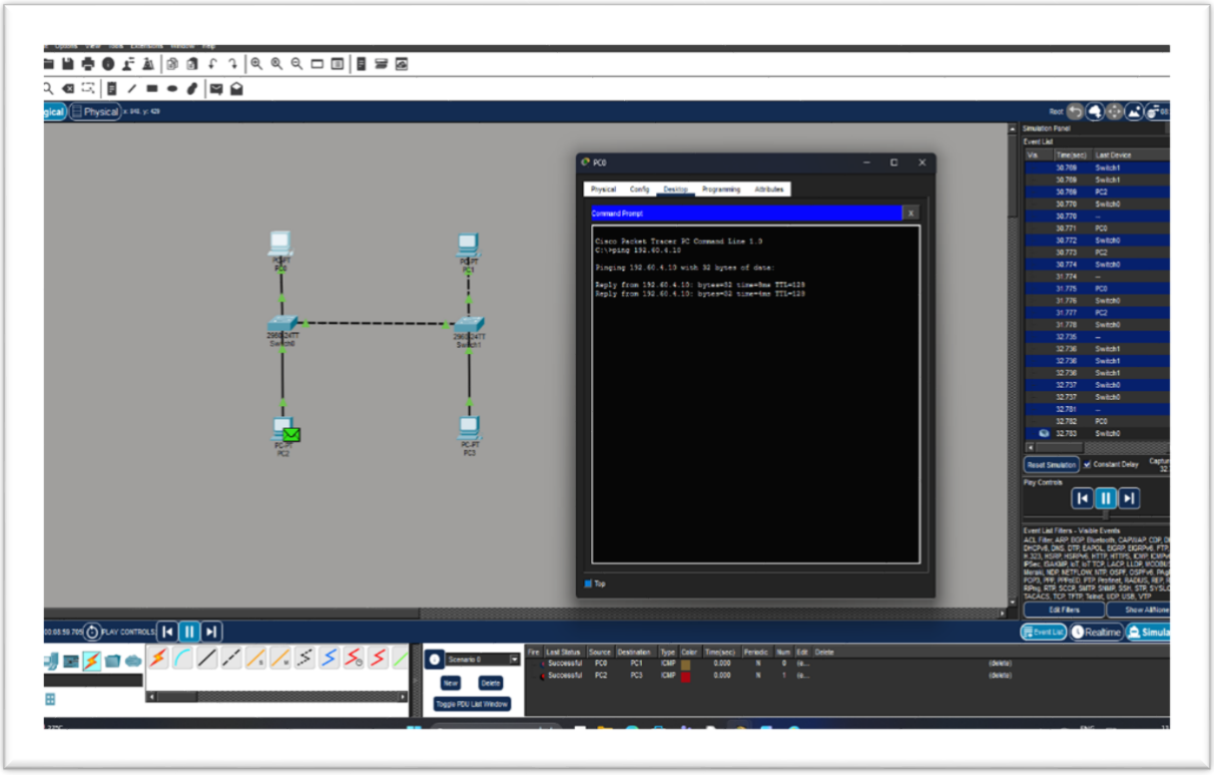
## Practical 2: Understanding Network Topologies

a. **AIM:** Create a star topology in Packet Tracer.

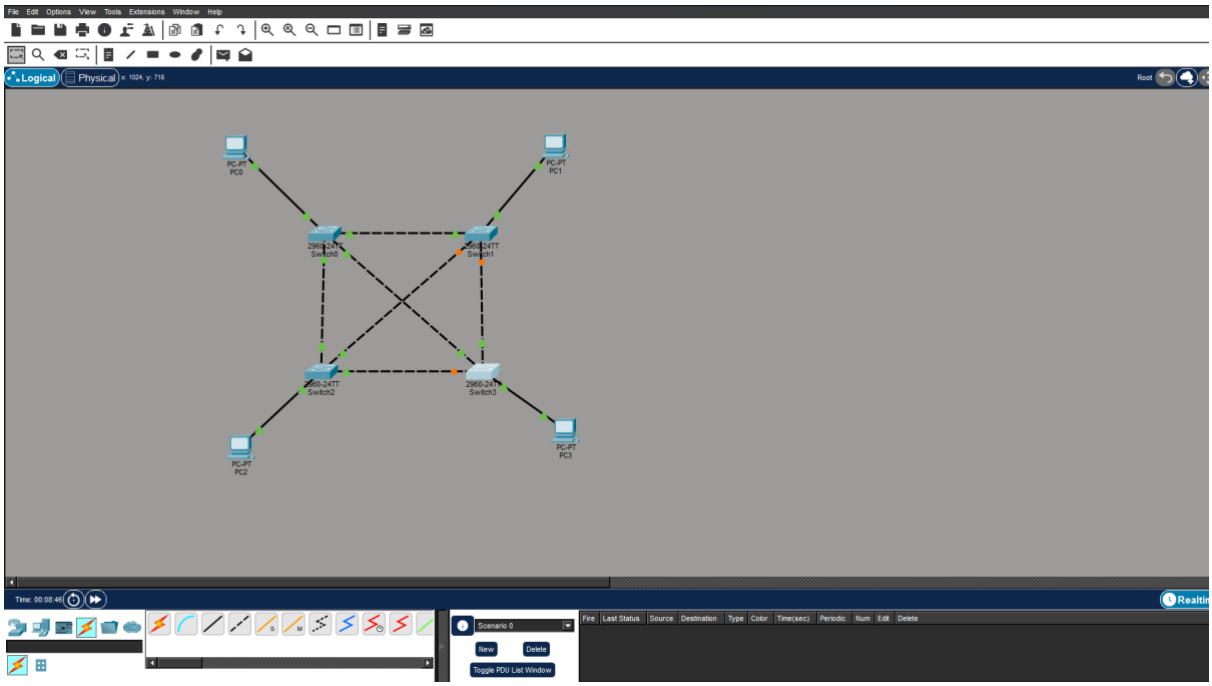


**b. AIM:** Simulate a bus topology and analyse traffic flow.

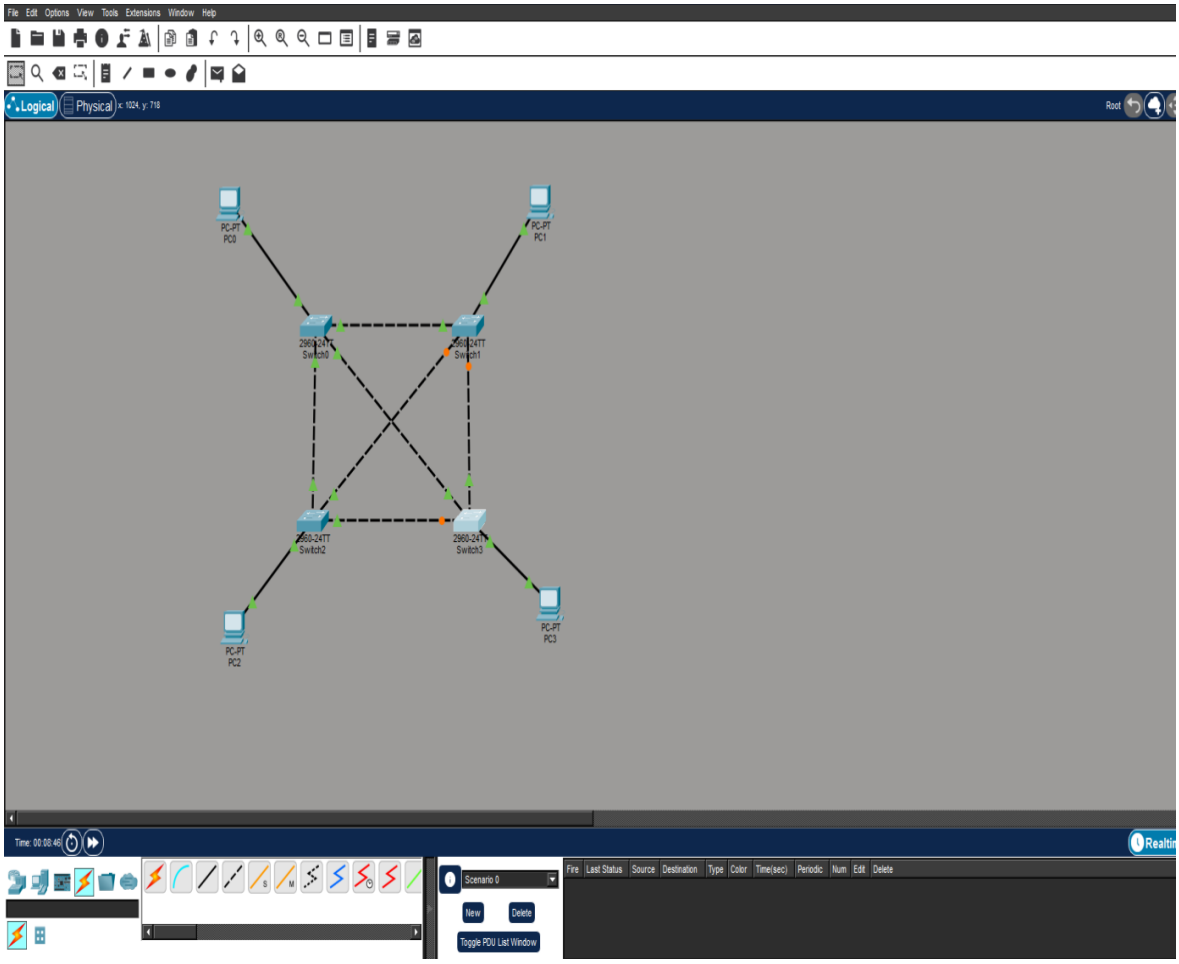




**c.** **AIM**: Demonstrate data communication in mesh topology.



**d. AIM**: Study of basic network command and Network configuration commands. Ipconfig, netstat, ARP etc.




```
acer PC Command Line 1.0
fig /all

 Connection:(default port)

tion-specific DNS Suffix..:
al Address.................: 0060.5CBD.C726
ocal IPv6 Address.........: ::
ress.......................: 0.0.0.0
 Mask......................: 0.0.0.0
t Gateway..................: 0.0.0.0
rvers......................: 0.0.0.0
ervers.....................: 0.0.0.0
 Client DUID...............: 00-01-00-01-0C-87-DA-DB-00-01-43-3B-BB-BB

net0 Connection:

tion-specific DNS Suffix..:
al Address.................: 0001.433B.BBBB
ocal IPv6 Address.........: FE80::201:43FF:FE3B:7596
ress.......................: 10.1.1.12
 Mask......................: 255.0.0.0
t Gateway..................: 0.0.0.0
rvers......................: 0.0.0.0
```

**Practical 03**

**Practical 3: Error Detection and Correction Techniques**

a. **AIM:** Simulate a data transmission scenario using parity bit error detection.

Q. Consider data transmitted by the sender is E, A, C, q the received bites by the receiver are 10001011, 10000010, 11100111, 11100000 each character data has been applied by a simple parity check detect using simple parity check detection method whether the recevied data has any error or not if you which is the character error.

Solution :-

E = 69 = 1000101
A = 65 = 1000001
C = 67 = 1000011
q = 113 = 1110001

Sender side

| Data | Binary Data | Parity | Data to be sent |
|---|---|---|---|
| E | 1000101 | 1 | 10001011 |
| A | 1000001 | 0 | 10000010 |
| C | 1000011 | 1 | 10000111 |
| q | 1110001 | 0 | 11100010 |

Receiver side

| Data | Send data | Send parity | Received parity | error |
|---|---|---|---|---|
| 10001011 | 10001011 | 1 | 1 | Yes |
| 10000010 | 10000000 | 0 | 0 | Yes |
| 11100111 | 10000111 | 1 | 1 | Yes |
| 11100000 | 11100010 | 0 | 1 | No |

error

b. **AIM:** Implement a CRC (Cyclic Redundancy Check) calculator and simulate bits errors.

CRC :-

Cyclic Redancy check.

- It is based on binary division.
- Total bits = message + Redantant bits
- It can find single bit burst, errors = polynominal length.
- It is the most commonly used error detection method.

Q. Data to be sent is 1010101010 and the polynomial redundancy bits are $x^4 + x^3 + 1$

A] Find the message sent by the sender.
B] Padded bits
C] If the data received by the receiver is 1010101010100100 has error or not.
D] Length of message.

Solution :-

$x^4 + x^3 + 1 = 1x^4 + 1x^3 + 0x^2 + 0x^1 + 1$

$= 11001 \longrightarrow$ divisor bits

$10 + 4 = 14 \longrightarrow$ size

polynominal highest degree.

Division = module 2 $\longrightarrow$ XOR

```
11001 | 1010101010000
        11001
        011000
         11001
        00001101 0
            11001
            00011000
              11001
              00001 0
```

101010101000 10

```
11001 | 10101010100100001 0
        11001
        011000
         11001
        00001101 0
            11001
            00011010
              11001
              00011000
                11001
                0000110
```

c. **AIM:** Demonstrate Simple check and 2D check mechanism.

## Simple Parity check :-

It is a base error detection technique in which extra bit falled parity bit is added to the data can take only bit error threre are 2 types of parity even parityhe parity is said so that the ones in the data ~~d~~ to even add parity the parity is said so that the number of one's in the data is add.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 |

Using simple parity check

| data | + | parity bit |
|---|---|---|

If is in data is add parity added is '1'
If is in data is even parity added is '0'

In given ex –

data

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 |

parity

Data Set

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 |

Q. Sender data E, A, C, q
Received data 10001011, 10000010, 11100111, 11100000
finds the error using 2D parity check.

Solution :-

Sender side

| Data | | | | | | | | Column parity |
|---|---|---|---|---|---|---|---|---|
| E | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| A | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| C | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| q | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 |
| Row parity | 0 | 1 | 1 | 0 | 1 | 1 | 0 | |

| | | | | | | | Co | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | | |
| 0 | 1 | 1 | 0 | 1 | 1 | 0 | | |
| ✓ | ✗ | ✗ | ✓ | ✓ | ✓ | ✗ | | |

**Practical 04**

**Practical 4: IP Addressing and Subnetting**

a. **AIM:** Convert between binary and decimal IPs

IP 205.16.37.39/27

| 27 | 5 |
|---|---|
| block / Network | host |

No of host - $2^5 = 32$

Network mask
11111111.11111111.11111111.11100000
255. 255. 255. 224

1st IP of the block:
255.255.255.224/27
AND
205.16.37.39/27

1st IP; 205.16.37.32/27

Last IP of the block:
205.16.37.63/27

Note: for the last IP consider host bits
for the first IP consider network bits

b. **AIM:** Subnet a class C IP address and assign subnets

IP 216.21.5.0

Sol: class C

Default mask
11111111.11111111.11111111.00000000

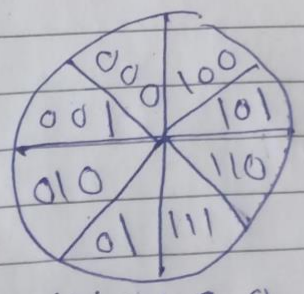
No of host = 30 = 1110 = 5 bits

subnet generator = 8 − 5 = 3

octet position = 255.255.255.11100000

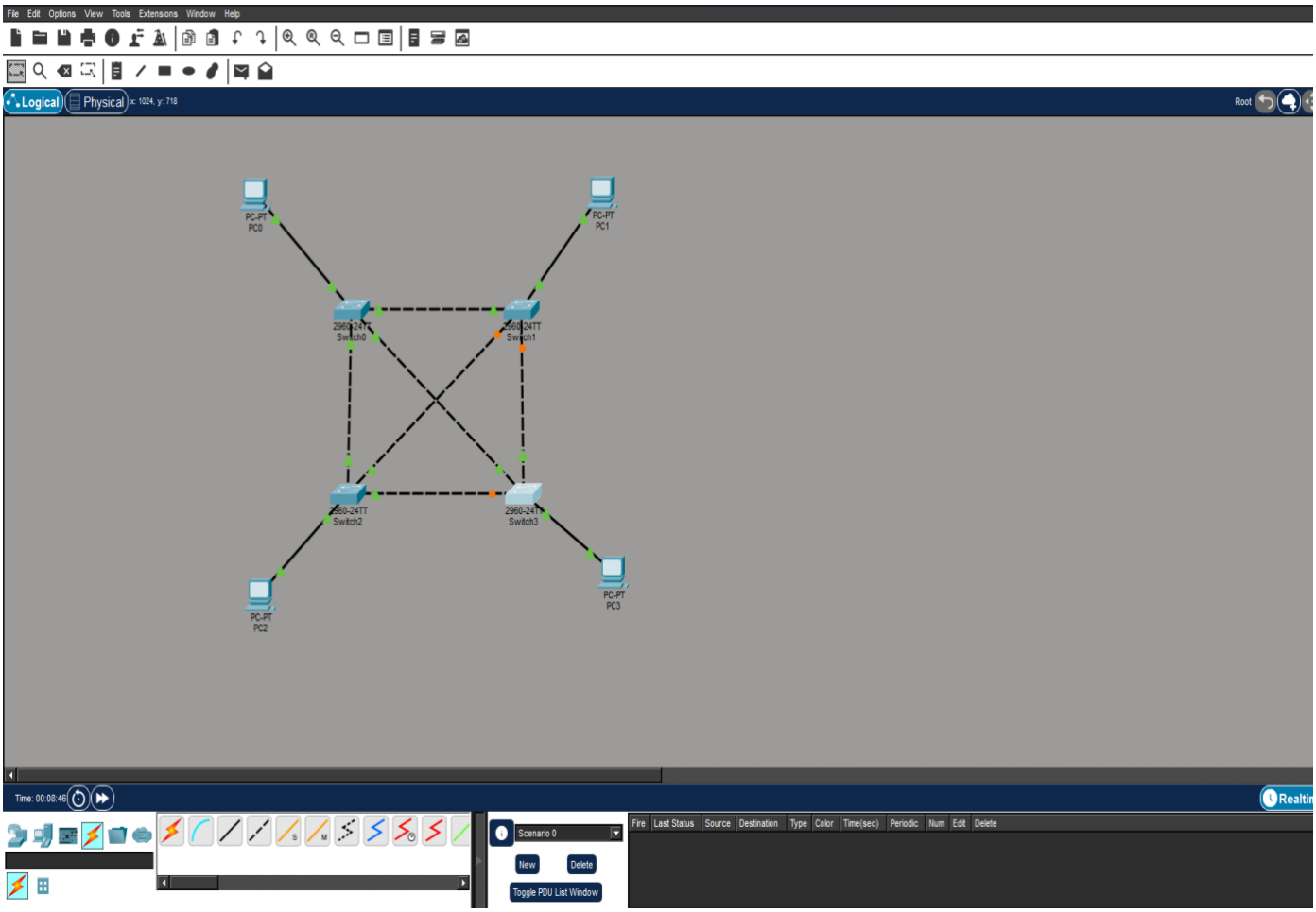
Network id : 216.21.5.0
Broadcast id : 216.21.5.254

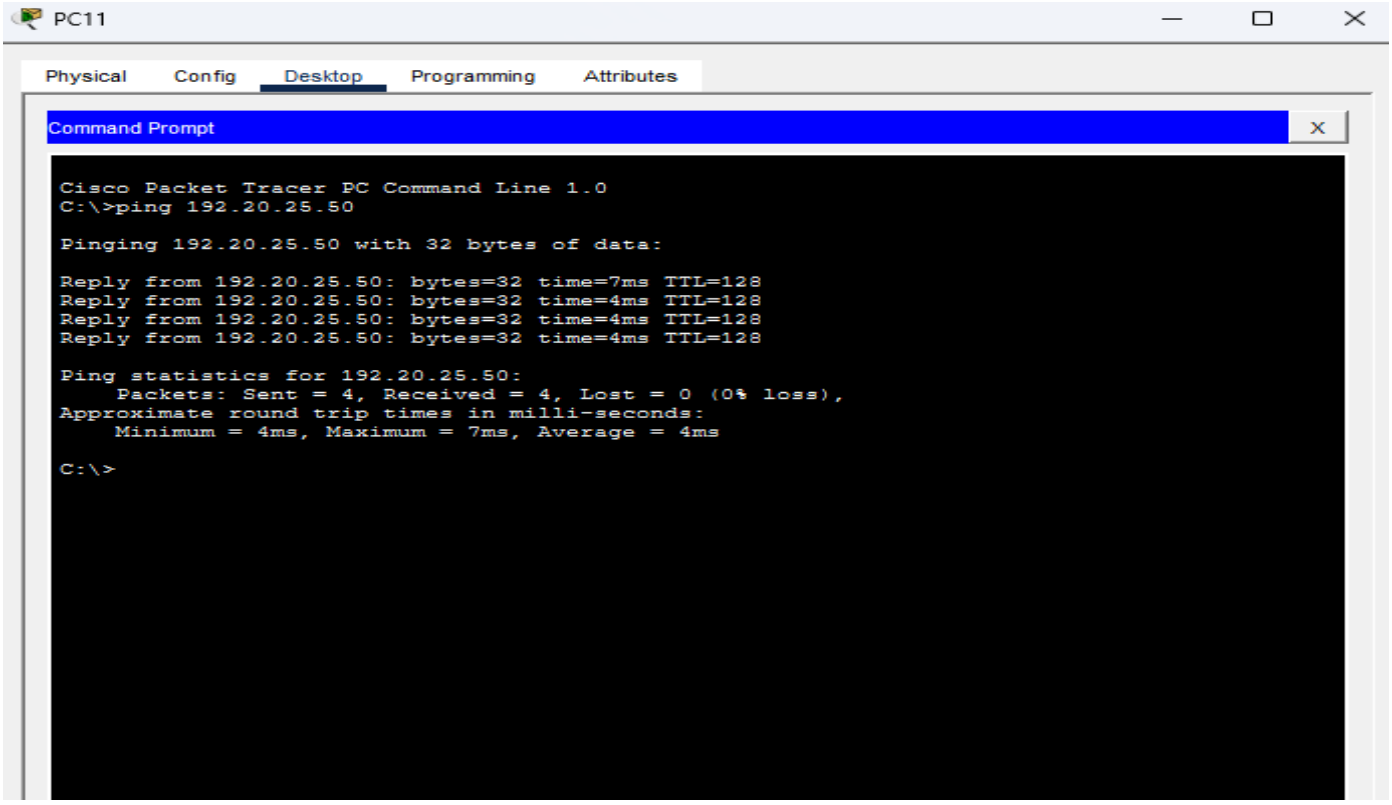
subnet group 1 : 216.21.5.0 to 216.21.5.31
subnet group 2 : 216.21.5.32 to 216.21.5.63
subnet group 3 : 216.21.5.64 to 216.21.5.95
subnet group 4 : 216.21.5.96 to 216.21.5.127
subnet group 5 : 216.21.5.128 to 216.21.5.159
subnet group 6 : 216.21.5.160 to 216.21.5.191
subnet group 7 : 216.21.5.192 to 216.21.5.223
subnet group 8 : 216.21.5.224 to 216.21.5.254




216.21.5.0

c. **AIM:** Configure IPs on Packet Tracer devices.



Router0

Physical Config CLI Attributes

IOS Command Line Interface

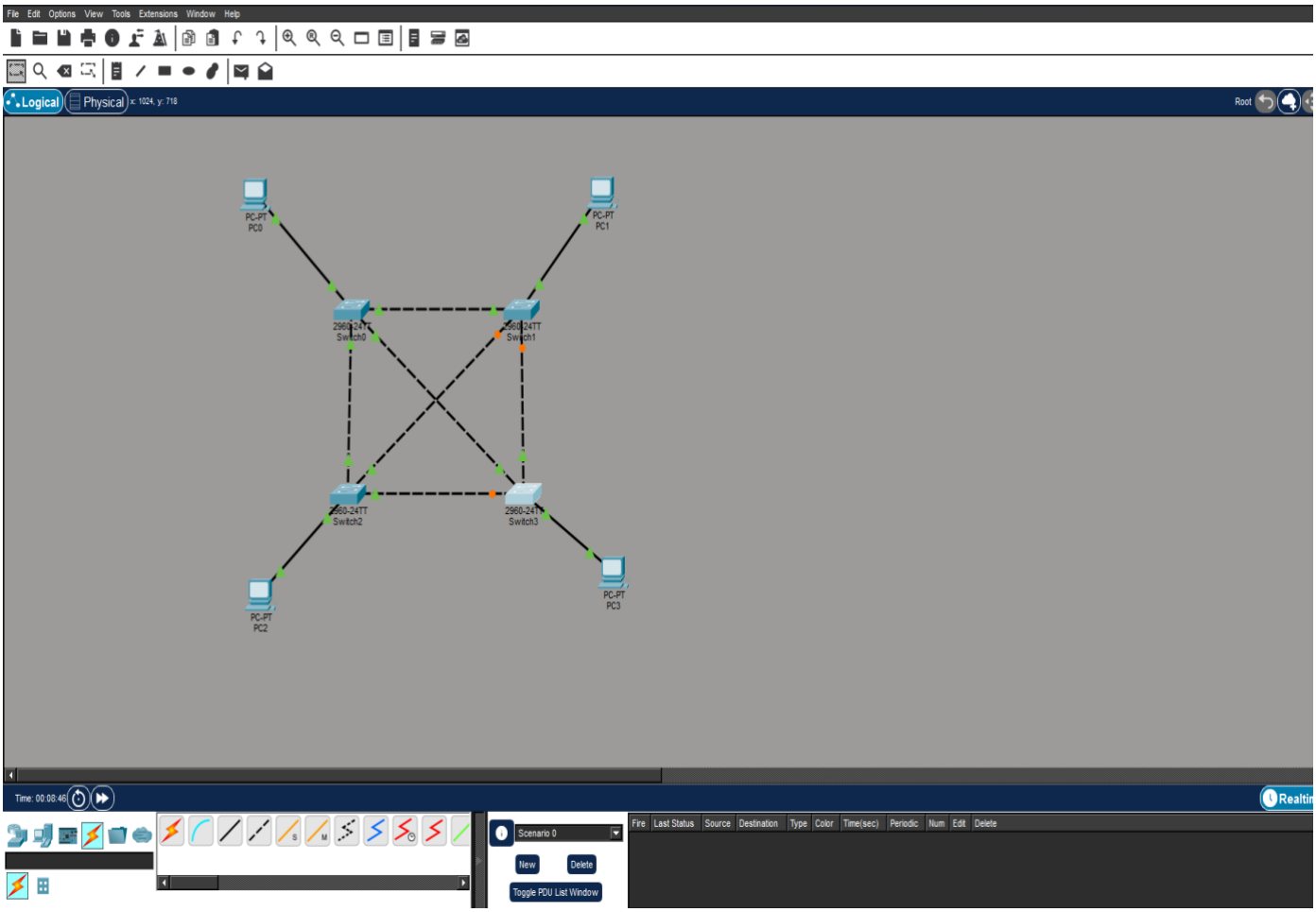
```
Router>en
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/1
Router(config-if)#ip address 192.168.10.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up

Router(config-if)#exit
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.20.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#exit
Router(config)#interface se0/0/0
Router(config-if)#ip address 192.168.50.1 255.255.255.0
Router(config-if)#no shutdown
```

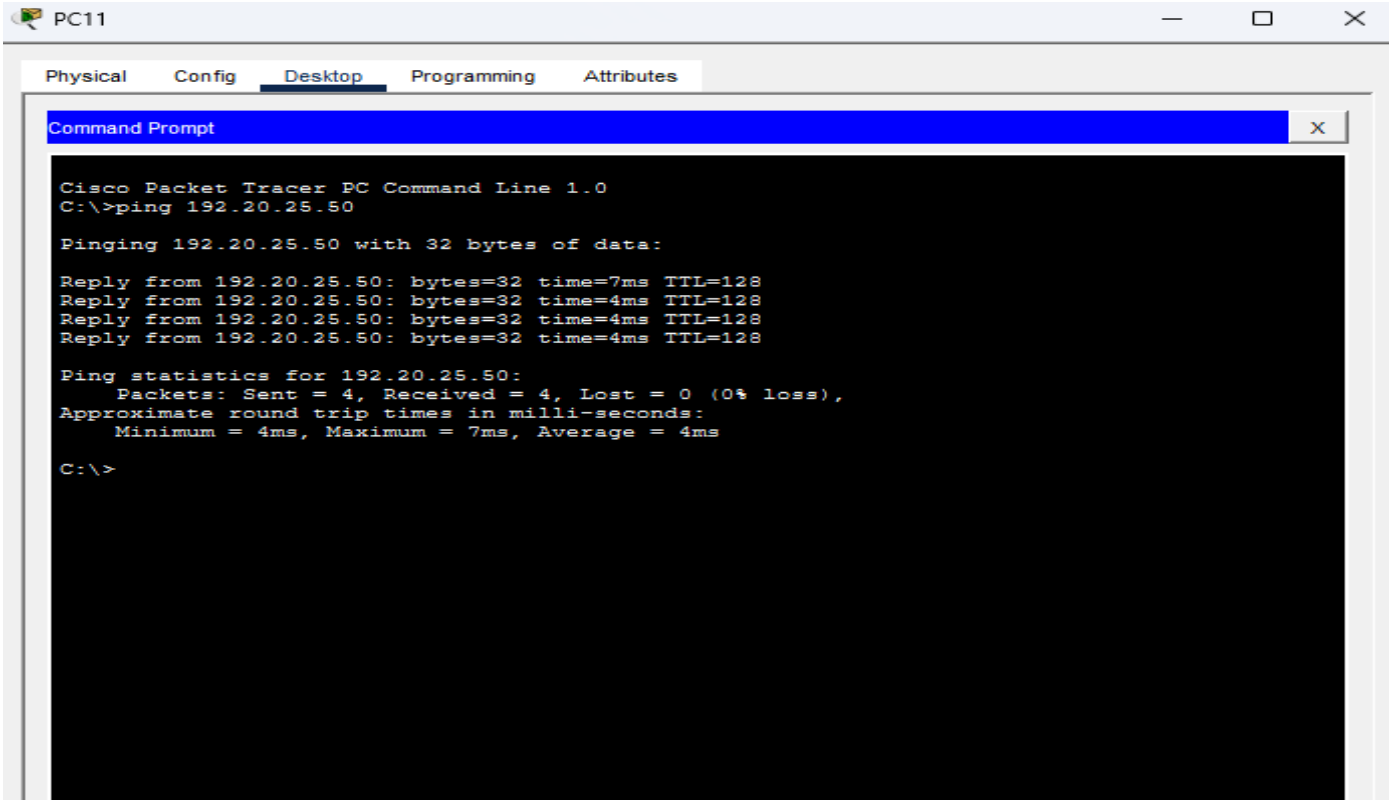
Ctrl+F6 to exit CLI focus

Copy Paste

Top

d. **AIM:** Validate connectivity using ping and trace route.

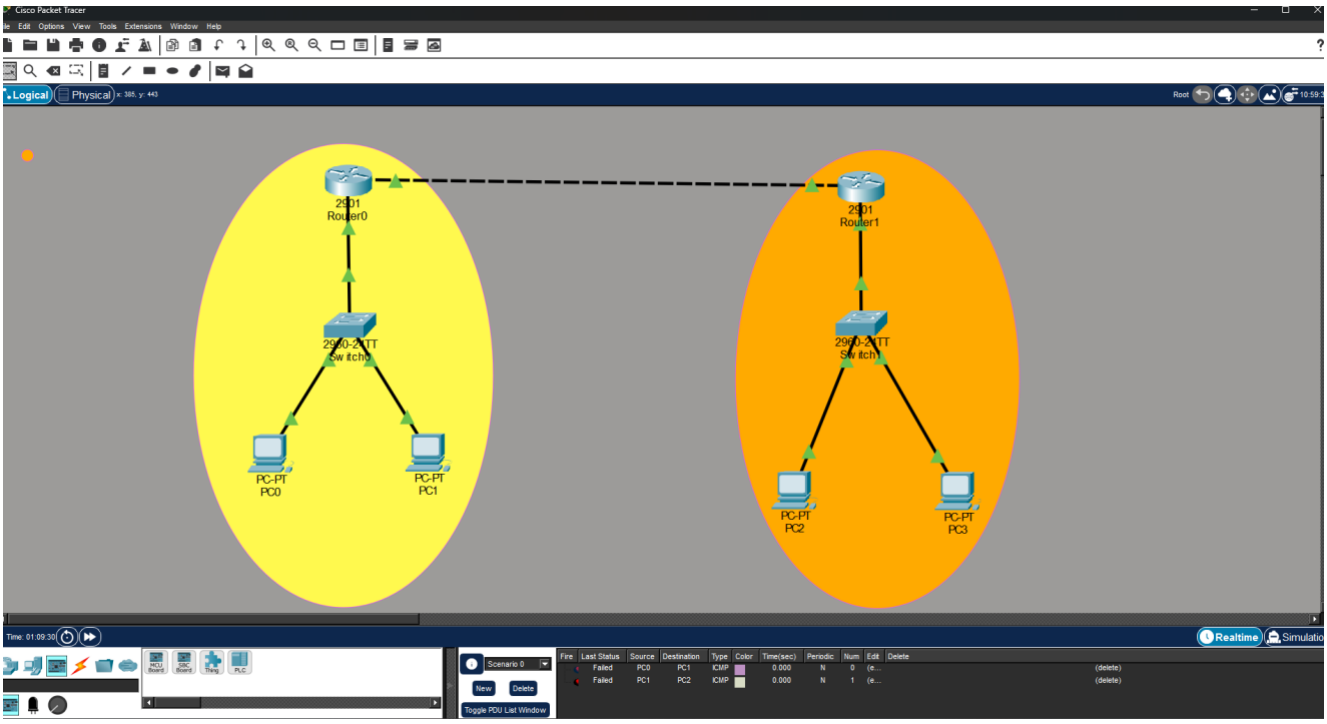


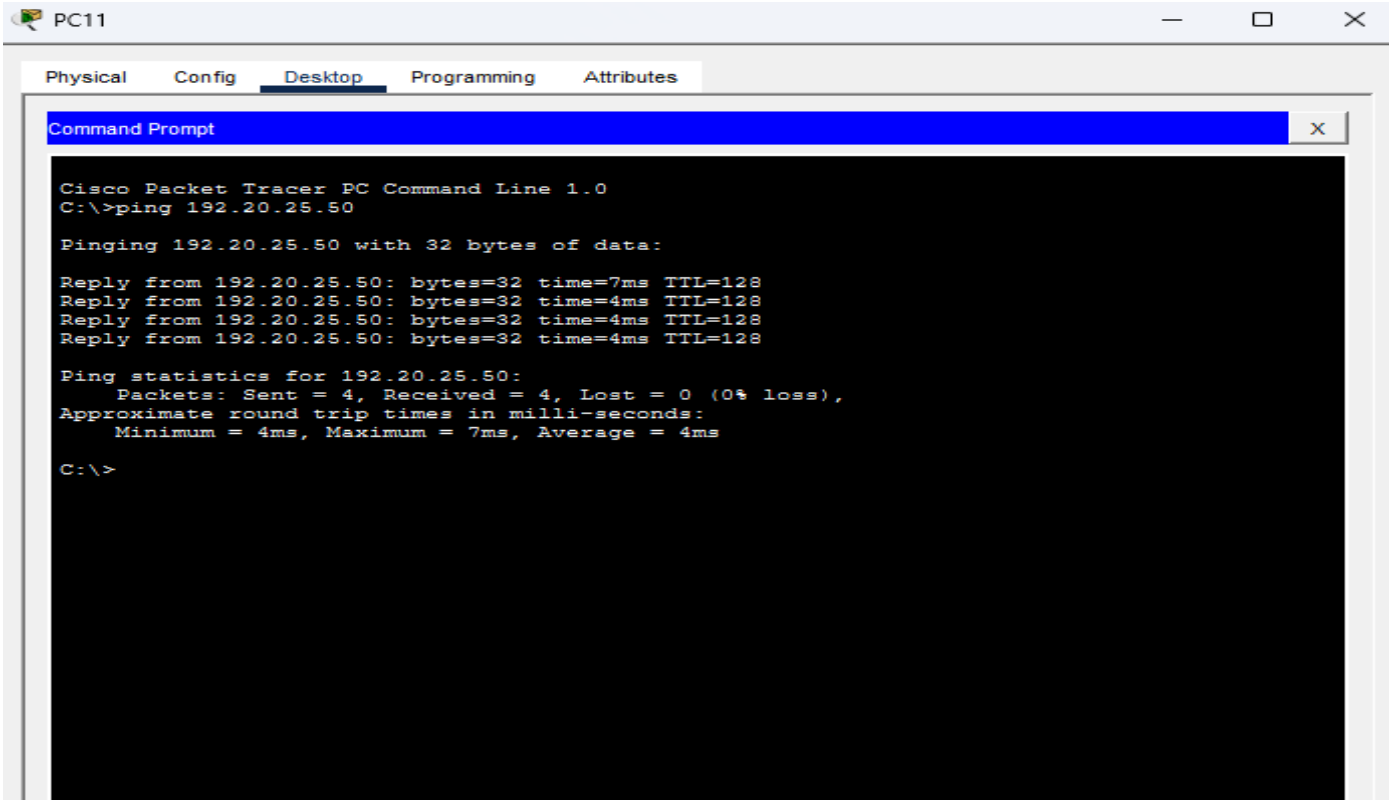



# Practical 05

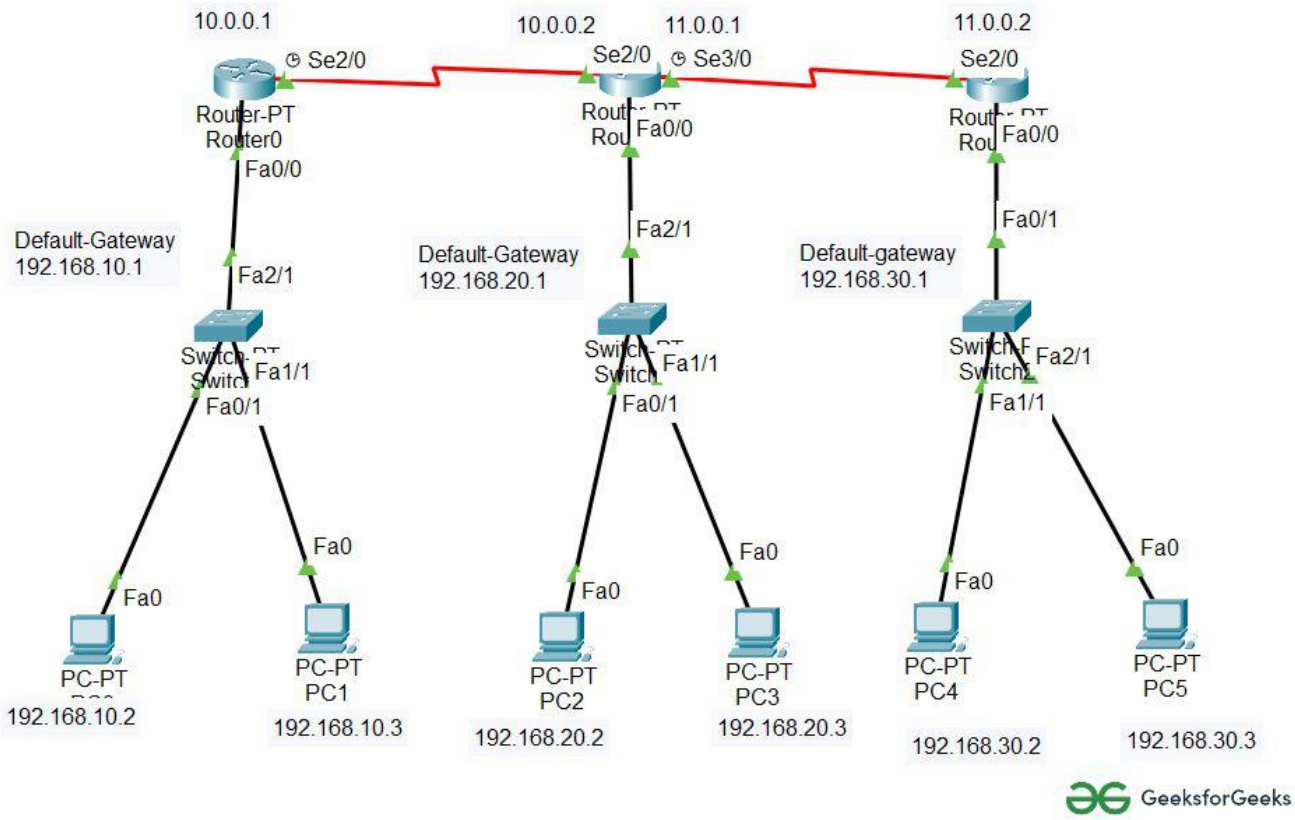
## Practical 5: Routing Protocols in Action

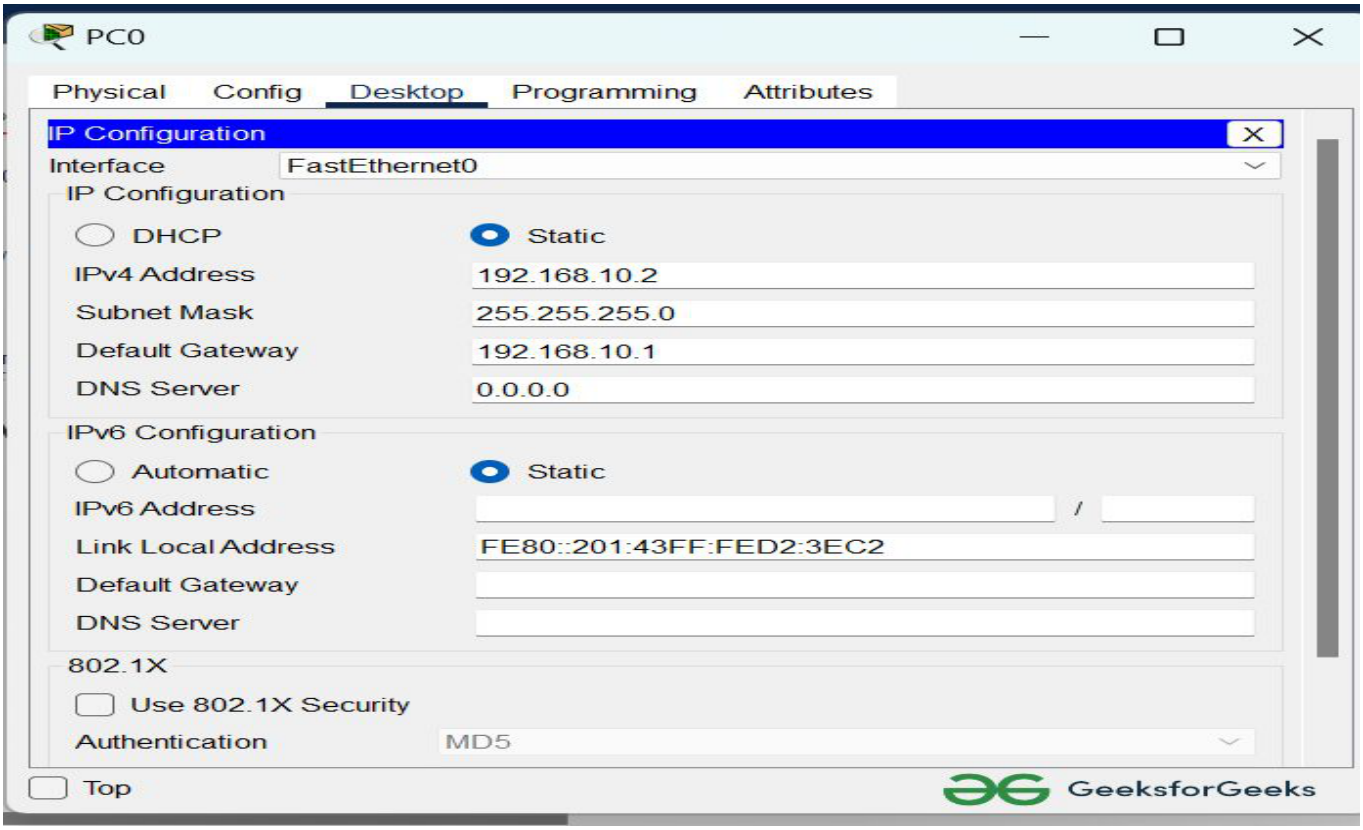
a. **AIM:** Implement static routing between 2 routers.



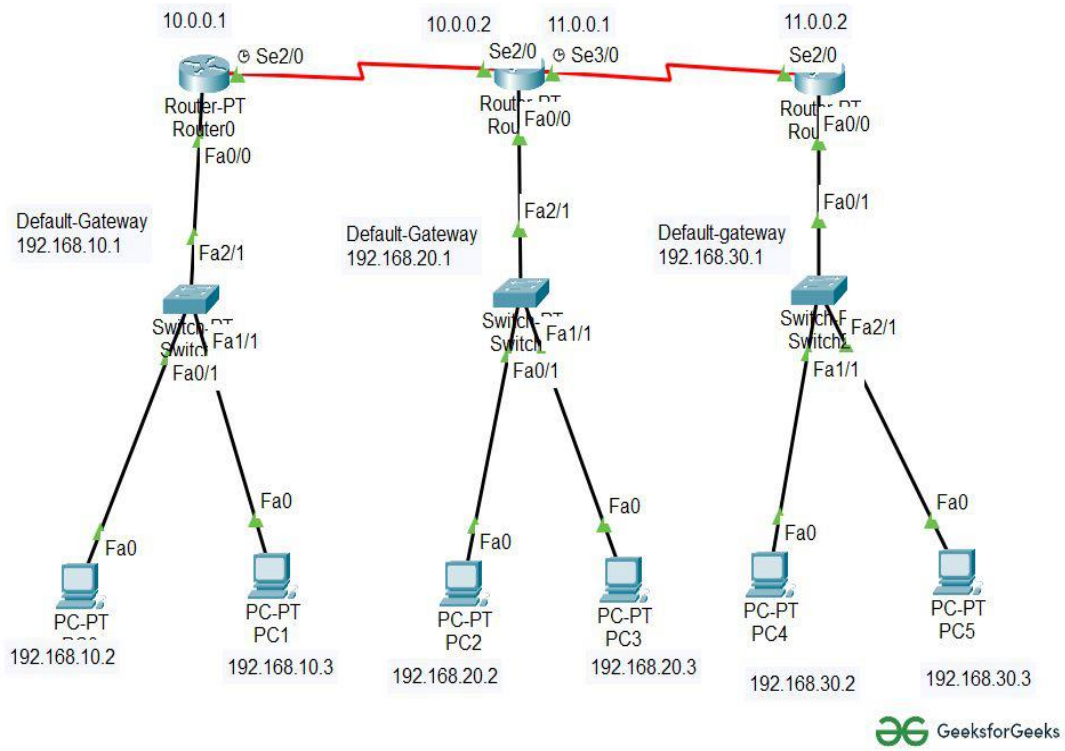



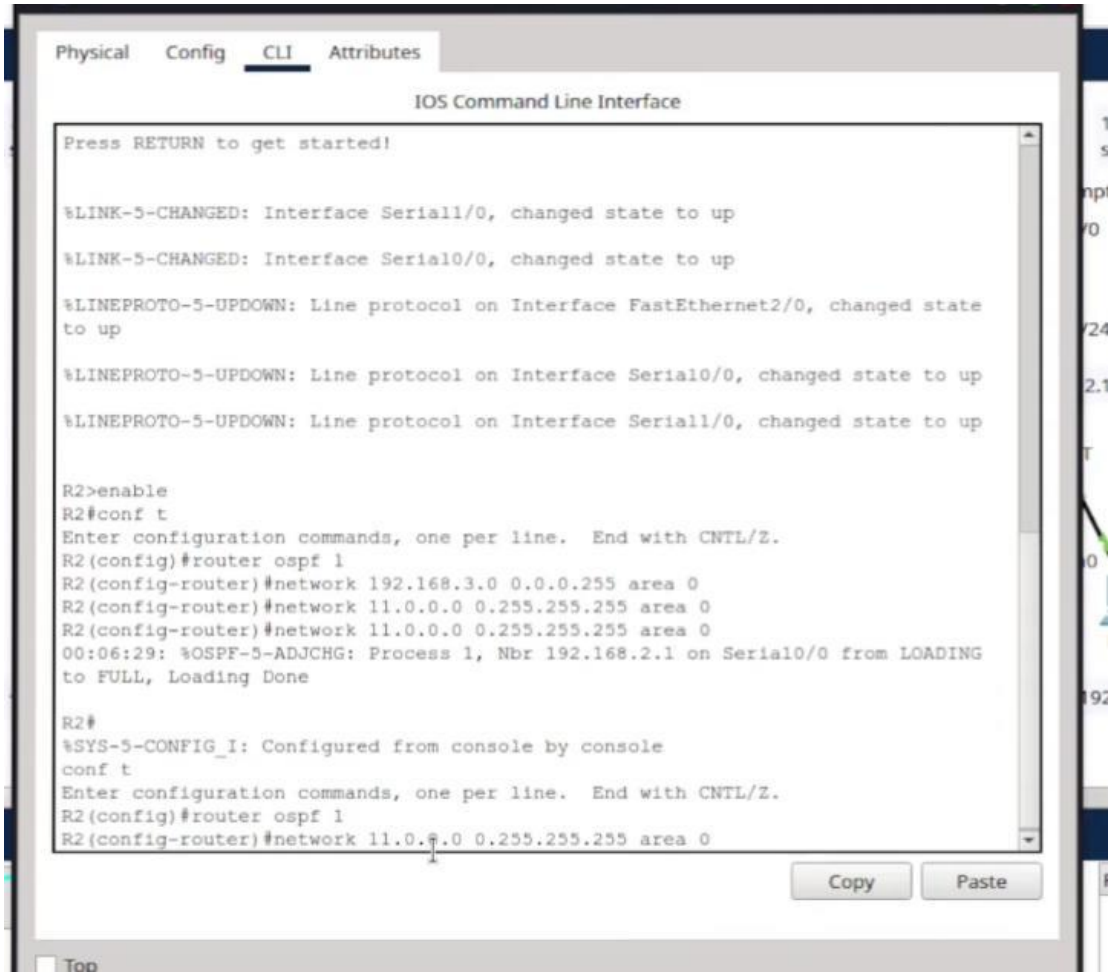
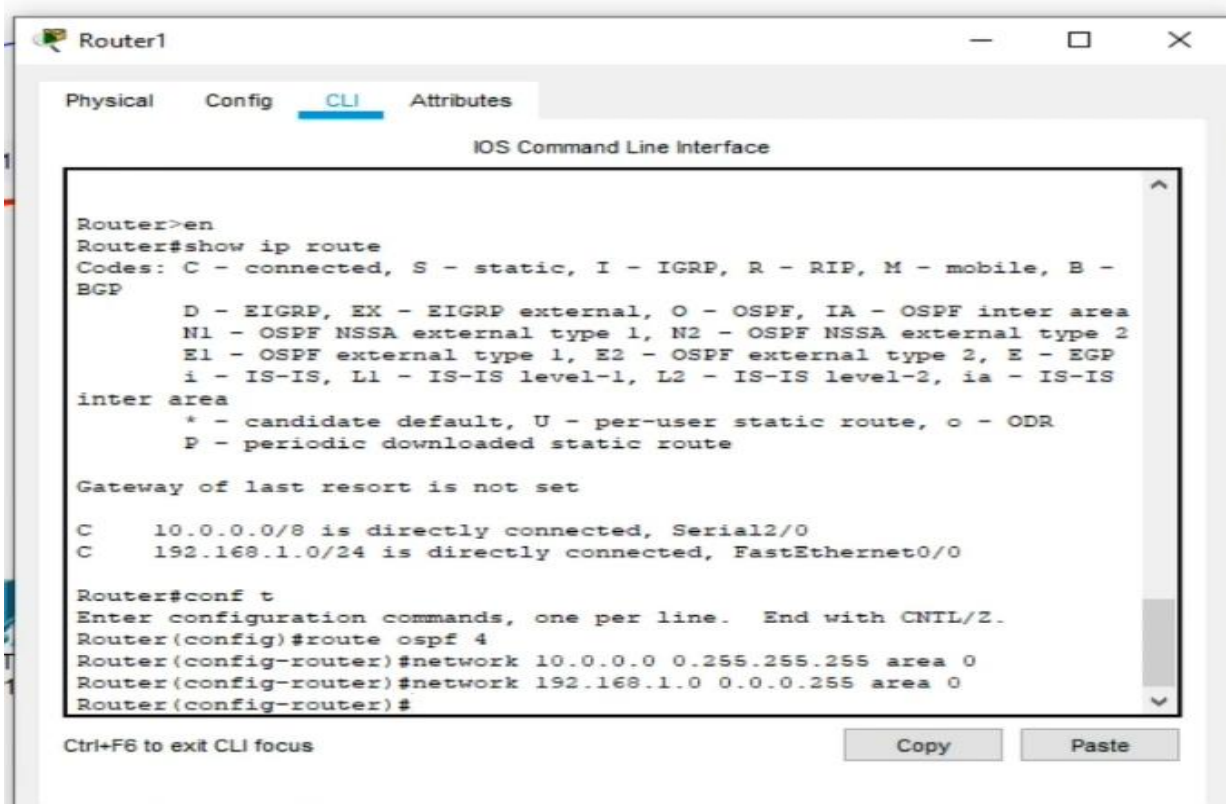
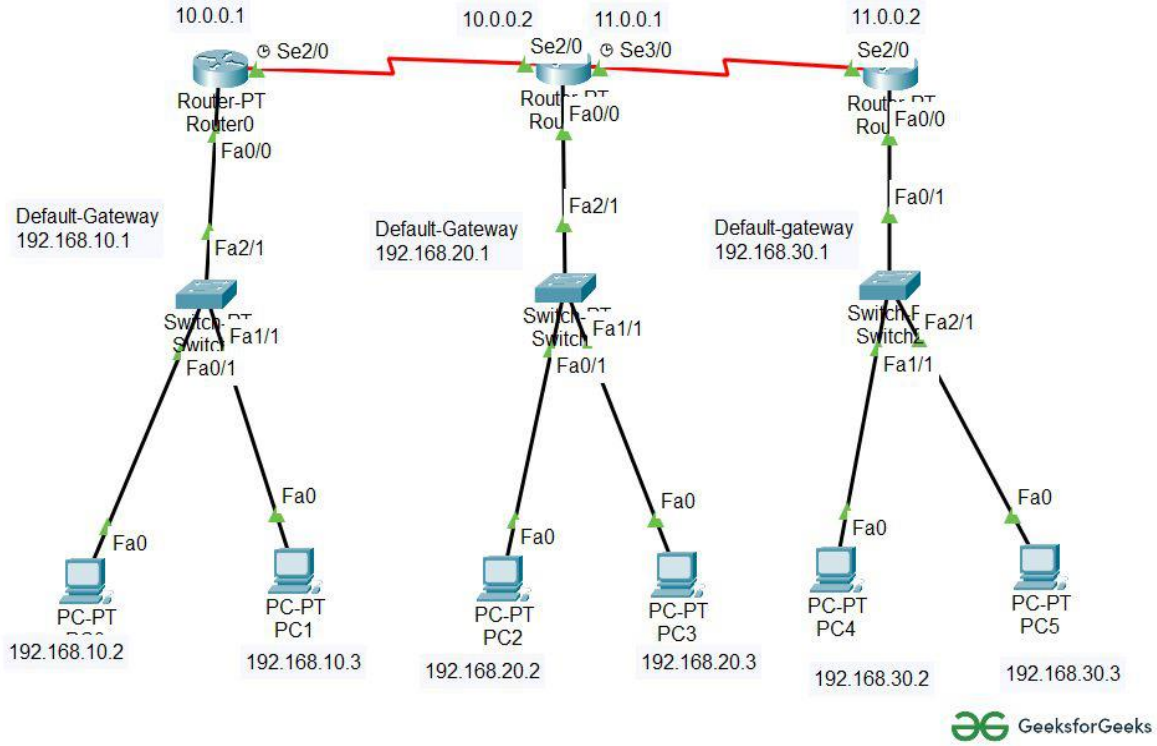
b. **AIM:** Configure RIP on a three-router topology.



c. **AIM:** Set up simple OSPF with area configuration.






d. **AIM:** Set up multiarea OSPF with area configuration.